

How to Play with a Child Holding Mythic Weapons

One SPS Case

Direct Excel Results

Moxian Qian

Helmholtz Institute Mainz (HIM)

GitHub: github.com/qxxmax/skillforpaper

Contents

01	Invoke and lock	Prompt, target paper, and C0–C4.
02	Search and screen	Query expansion, candidate pool, and Dijkstra ranking.
03	Read and source	Source table, problem and method, results and limits.
04	Lineage and bounds	Citation lineage, claims, evidence, and boundaries.
05	Backup	Run scale, elapsed time, tokens, and unsupported conclusions.

Invoke

Prompt	Use play-the-toy-with-children for Part 1 literature research.
Verbal clue	Moxian Qian's recent stochastic path sampler paper.
Requirements	Lock the target; expand repeatedly; read selected papers; return to original sources; preserve sources, boundaries, and open questions.
Outputs	source table; reading notes; claim/boundary ledger; lineage; gaps; report.



Target Lock

Field	Verified result
Title	Stochastic Path Sampler For Lattice Field Theory
Authors	Shiyang Chen; Moxian Qian; Gert Aarts; Biagio Lucini; Kai Zhou
arXiv	2606.13790v1
Date	2026-06-11
Primary sources	arXiv page and downloaded PDF
Order	Lock identity before expanding citation and method lineages.

Level	Promotion condition	What it supports
C0	Search result	Candidate.
C1	Title, authors, date, ID	Identity.
C2	Working page/PDF	Object exists.
C3	Original context	Lineage.
C4	Claim + exact locator	Report-ready.

Search Expansion

	A	B	C	D	E	F
1	query_id	family	source	query	facet	parent_evidence
2	Q01	root	OpenAlex	"Stochastic Path Sampler For Lattice Field Theory"	identity	root title
3	Q02	root	Crossref	"Stochastic Path Sampler For Lattice Field Theory"	identity	root title
4	Q03	root	arXiv	ti:"Stochastic Path Sampler For Lattice Field Theory"	identity	root title
5	Q04	backward	root_bibliography	all live root references	lineage	root TeX
6	Q05	forward	OpenAlex	works citing arXiv:2606.13790	forward	root arXiv ID
7	Q06	author	OpenAlex	"Shiyang Chen" lattice sampler	author	root title page
8	Q07	author	OpenAlex	"Moxian Qian" lattice sampler	author	root title page
9	Q08	author	OpenAlex	"Gert Aarts" lattice sampler	author	root title page
10	Q09	author	OpenAlex	"Biagio Lucini" lattice sampler	author	root title page



Every query retains its **family, source, facet, and parent evidence.**

Candidate Screening

	A	B
1	candidate_id	title
2	arxiv:2606.13790	Stochastic Path Sampler For Lattice Field Theory
3	arxiv:2502.05504	Physics-Conditioned Diffusion Models for Lattice Gauge Theory
4	arxiv:2409.15937	Numerical determination of the width and shape of the effective string using Stochastic Normalizing Flows
5	arxiv:2412.19109	Stochastic normalizing flows for Effective String Theory
6	arxiv:2210.03139	Stochastic normalizing flows for lattice field theory

	M	N
1	score	decision
2		121 include
3		20 include
4		19 include
5		18 include
6		18 include

The final pool has **593 candidates**. Screening prioritizes reading; a candidate is not yet evidence.

Dijkstra Ranking

	A	B	C	D	E
1	metric	relevance_only	candidate_dijkstra	final_with_gap_closure	interpretation
2	selected papers	30	30	37	fixed reading budget for the first two arms
3	declared facets covered	7	7	10	Dijkstra did not increase coarse facet count
4	method groups covered	8	8	8	coverage equality does not imply identical papers
5	exact root-bibliography papers	19	21	21	candidate Dijkstra retained two more root ancestors
6	mean heuristic screen score	16.7	16.067	not comparable	graph navigation traded a little keyword score for paths
7	selection overlap	20	20	35	20/30 papers overlap; ten are replaced

With the same 30-paper budget, Dijkstra retained **two additional direct references of the target paper**; coarse facets and method groups did not increase.

Source Table

	A	B	C
1	paper_id	arxiv_id	title
2	P001	2606.1379	Stochastic Path Sampler For Lattice Field Theory
3	P002	2111.15141	Path Integral Sampler: a stochastic control approach for sampling
4	P003	2302.13834	Denoising Diffusion Samplers
5	P004	2307.0105	Transport meets Variational Inference: Controlled Monte Carlo Diffusions
6	P005	2410.02711	NETS: A Non-Equilibrium Transport Sampler
7	P006	2210.03139	Stochastic normalizing flows for lattice field theory
8	P007	2201.08862	Stochastic normalizing flows as non-equilibrium transformations

	G	H
1	source_relation	source_url
2	root	https://arxiv.org/abs/2606.13790
3	direct_bibliography	https://arxiv.org/abs/2111.15141
4	direct_bibliography	https://arxiv.org/abs/2302.13834
5	direct_bibliography	https://arxiv.org/abs/2307.01050
6	direct_bibliography	https://arxiv.org/abs/2410.02711
7	direct_bibliography	https://arxiv.org/abs/2210.03139
8	direct_bibliography	https://arxiv.org/abs/2201.08862

The final run selected and read **37 full texts**; each row retains identity, relation, and primary URL.

Problem and Method

	I	J
1	problem	problem_anchor
2	Data-free sampling of an unnormalized lattice target near criticality, where ordinary chains develop long autocorrelation times.	PDF p.1 abstract
3	Construct an efficient sampler for an unnormalized target by controlling a stochastic path rather than fitting a terminal density alone.	PDF p.1 abstract
	K	L
1	method	method_anchor
2	Learn paired forward and auxiliary backward Langevin paths by minimizing a path-space variational free energy / entropy-production upper bound.	PDF pp.4-8 Sec. 2
3	Formulate sampling as a Schrodinger-bridge / stochastic optimal-control problem and learn a neural control for a reference diffusion.	PDF pp.4-8 method

Every problem and method statement carries a PDF page or section anchor.

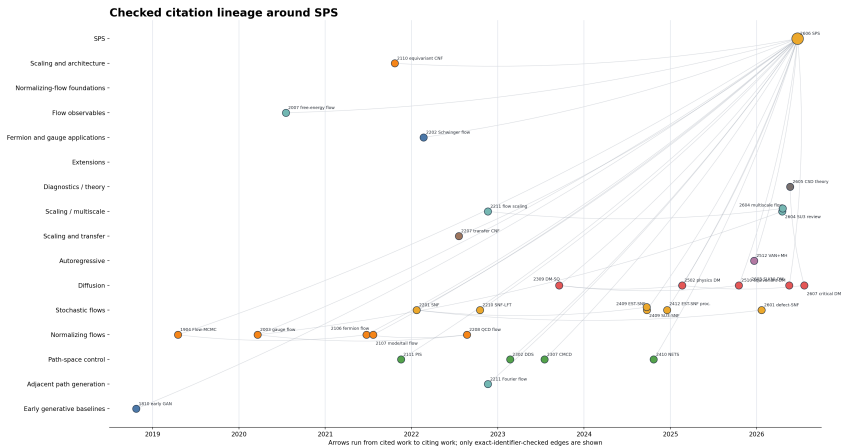
Results and Limits

	M	N	O
1	correction_exactness	result	result_anchor
2	Independent path proposals are corrected with an extended-space independence Metropolis-Hastings step;	Corrected SPS observables agree with HMC in the tested 2D ϕ^4 systems and show shorter autocorrelation near the pseudocritical region.	PDF pp.12-18 Sec. 3, Tables 1-4
3	Girsanov path likelihood ratios provide importance weights; weighted estimates retain the path-measure correction.	The learned path sampler is competitive with the reported MCMC, SMC and normalizing-flow baselines on its benchmark suite.	PDF pp.9-18 experiments

	P	Q
1	limitation	limitation_anchor
2	Uncorrected tails and multimodal regions deviate; finite network capacity, terminal diffusion and support coverage remain practical limits, and the reported timing is hardware/setup specific.	PDF pp.12-13 Secs. 3.1-3.2; pp.19-21 discussion
3	Training overhead, time discretization and a suboptimal learned control can produce weight variance and residual bias if correction is omitted.	PDF pp.19-20 discussion

Exactness, result, limitation, and their original anchors remain in the same row.

Citation Lineage



The horizontal axis is year, the vertical axis is method family, and arrows run from cited work to citing work.

Claims and Boundaries

	A	B	C	D	E	F
1	claim_id	claim	allowed_short_sentence	evidence_ids	boundary	status
2	C01	SPS learns paired forward and auxiliary backward stochastic paths with a path-space variational objective.	SPS learns a low-irreversibility path between prior and target.	E-260613790-M	This is the SPS v1 formulation; do not replace it with terminal-density KL training.	confirmed
3	C02	SPS follows learned independent path proposals with extended-space independence Metropolis-Hastings.	The learned SPS proposal is followed by an exactness gate.	E-260613790-M-E-260613790-R	Do not call uncorrected SPS samples exact.	confirmed
4	C03	The nearest method lineage combines path-space stochastic control, diffusion samplers and nonequilibrium transport.	SPS sits at the intersection of path-space control and nonequilibrium transport.	E-211115141-M-E-230213834-M-E-230701090-M-E-241002711-M-E-260613790-M	This is a checked synthesis, not a claim that the algorithms are identical.	confirmed
5	C04	The lattice lineage repeatedly pairs learned proposals with Metropolis, reweighting or Jarzynski correction.	Learned lattice proposals are repeatedly paired with an explicit correction mechanism.	E-190412072-M-E-200306413-M-E-210605934-M-E-220108862-M-E-220803832-M	Shared correction logic does not imply equal scaling or performance.	confirmed
6	C05	Diffusion models do not automatically remove critical slowing; architecture and locality control the outcome in the checked theory study.	Diffusion helps only when architecture and physics match the slow modes.	E-260512597-R-E-260512597-L-E-260708505-L	The quadratic-to-logarithmic result is controlled in the Gaussian Q(n to infinity) setting, not universally.	confirmed
7	C06	Cross-volume scores, coarse-to-fine factorization and localized defect maps are distinct scale-transfer routes.	Transfer, multiscale structure and locality are three distinct routes to larger lattices.	E-260708505-R-E-260410209-R-E-260120708-R	Each result is benchmark-specific; production-QCD scaling is not established.	confirmed
8	C07	A fair efficiency claim must include training, generation, correction and observable-level uncertainty.	Efficiency means physical error per total compute, not a single ESS number.	E-221107541-R-E-221107541-L-E-190412072-L-E-260410209-R-E-240918861-L	The literature does not yet supply one uniform cross-paper cost benchmark.	confirmed
9	C08	Aggregate loss or ESS can hide support, tail, multimodal and infrared errors.	A sampler can look good globally and still miss the modes that matter.	E-210700734-R-E-210700734-L-E-260708505-R-E-260613790-L	Diagnostics expose risk; they do not by themselves correct it.	confirmed
10	C09	Realistic lattice-QCD evidence remains sparse despite several gauge, fermion and 4D SU(3) demonstrations.	The field is active, but production-scale lattice-QCD evidence is still sparse.	E-260412416-R-E-260412416-L-E-220803832-L-E-260120708-R	Do not describe these methods as mature at production QCD scale.	confirmed
11	C10	The defensible SPS follow-ups close correctness, scaling and physics-cost loops together.	The next SPS step is to close correctness, scaling and physics-cost loops together.	E-260613790-L-E-210700734-L-E-260512597-L-E-260708505-L-E-260410209-L-E-210605934-L	These are literature-grounded directions, not SPS results already achieved.	confirmed

Claim, short sentence, evidence ID, boundary, and status live in one table.

Backup: Run Record

Recorded item	Final run
candidate graph equal reading budget selection effect after gap closure	3,300 nodes / 8,187 edges; local Dijkstra took about 0.11 s. relevance-only: 30 papers; Dijkstra: 30 papers; overlap: 20. exact root-bibliography papers: 19 → 21. 37 full texts; 10 facets; 185 evidence entries; 199 checked direct-citation relations.
complete goal not supported	1,012,083 tokens; 4,039 s; 19/19 validation gates PASS. Per-arm token/time was not measured; a Dijkstra path is navigation metadata, not scientific evidence.

The algorithm decides where to look first; original sources and evidence gates decide what can be written.